

Chapter 2 · Lesson 1

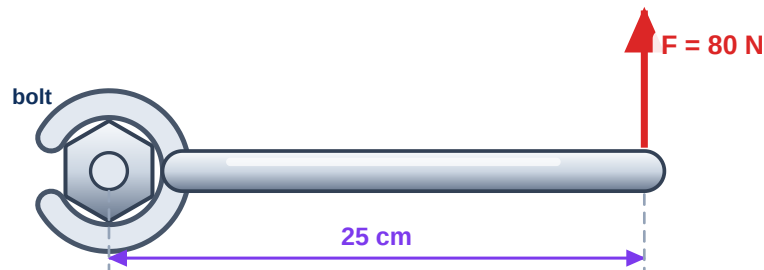
Moment of a Force

The turning effect of a force about a point · perpendicular and inclined forces · the moment arm · the net moment

Sign convention: anticlockwise (+) · clockwise (-)

1

A mechanic tightens a bolt using a spanner of length 25 cm. He applies a force of 80 N at the end of the spanner, perpendicular to it, as shown in the figure:

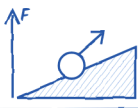


I The moment of the force about the axis of the bolt equals

- a) 320 N·m b) 20 N·m c) 2000 N·m d) 3.2 N·m

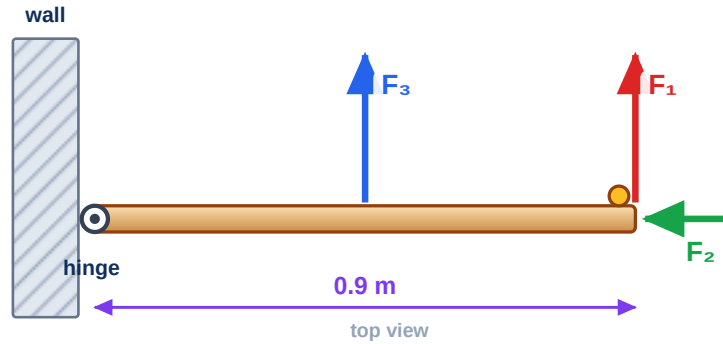
II If he applies the same force at the middle of the spanner, the moment becomes

- a) 40 N·m b) 20 N·m c) 5 N·m d) 10 N·m



2

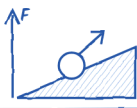
The figure shows the top view of a door of width 0.9 m hinged at one side. Three forces F_1 , F_2 and F_3 act on the door, each separately:



Which of the three forces has no moment about the hinge? Explain your answer.

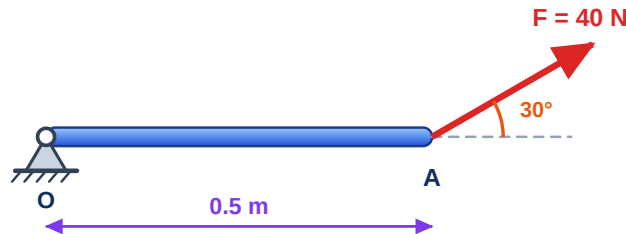
If $F_1 = F_3 = 30 \text{ N}$, the ratio between their moments $M_1 : M_3$ equals

a) 1 : 2 b) 2 : 1 c) 1 : 1 d) 4 : 1



3

A light rod OA of length 0.5 m can rotate about a pivot at O. A force of 40 N acts at the end A, making an angle of 30° with the rod as shown:



The moment arm of the force about O (the perpendicular distance from O to the line of action) equals

- a) 0.5 m b) 0.43 m c) 0.87 m d) 0.25 m

The moment of the force about O equals

- a) +10 N·m b) +17.3 N·m c) +20 N·m d) -10 N·m

4

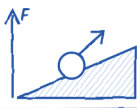
Choose the correct answer:

The SI unit of the moment of a force is

- a) N b) N/m c) J d) N·m

The moment of a force about a point equals zero when

- a) the force is perpendicular to the arm
b) the line of action of the force passes through the point
c) the force is very large
d) the arm is very long

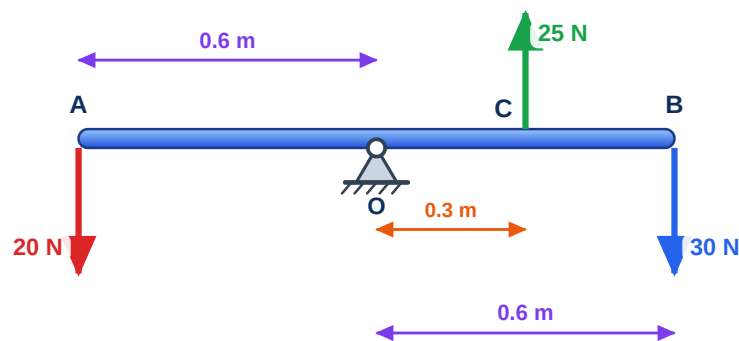


Door handles are fixed at the edge farthest from the hinges in order to

- a) increase the force needed to open the door
- b) decrease the moment of the force
- c) increase the moment arm, so a smaller force gives the needed moment
- d) make the door lighter

5

A light rod AB of length 1.2 m is pivoted at its midpoint O. Three forces act on it as shown: 20 N downwards at A, 30 N downwards at B, and 25 N upwards at C, where OC = 0.3 m:

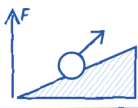


The moment of the 30 N force about O equals

- a) +18 N·m
- b) -9 N·m
- c) -18 N·m
- d) +9 N·m



Calculate the net moment about O, and state the direction in which the rod starts to rotate.



6

A force $F = 60 \text{ N}$ acts at point P of a rigid body that can rotate about O, where $OP = 0.5 \text{ m}$. The magnitude of the moment of F about O is $15 \text{ N}\cdot\text{m}$:



The angle between the force and the line OP could be

a) 30° only

b) 60° only

c) 90°

d) 30° or 150°



Find the smallest force that, acting at P, gives the same moment, and state its direction.

7

A stuck nut needs a moment of $36 \text{ N}\cdot\text{m}$ to loosen it. The only handle available is bent, and a force can be applied only at its end P, which is 0.3 m horizontally and 0.4 m vertically from the centre O of the nut, as shown:



Find the minimum force needed at P, and state its direction.



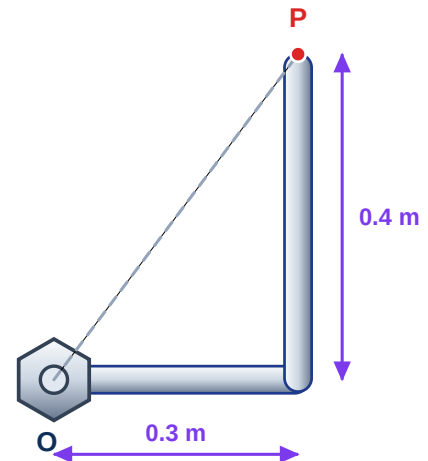
If the force at P must be horizontal, its magnitude is

a) 90 N

b) 72 N

c) 120 N

d) 60 N



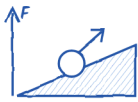
If the force at P must be vertical, its magnitude is

a) 72 N

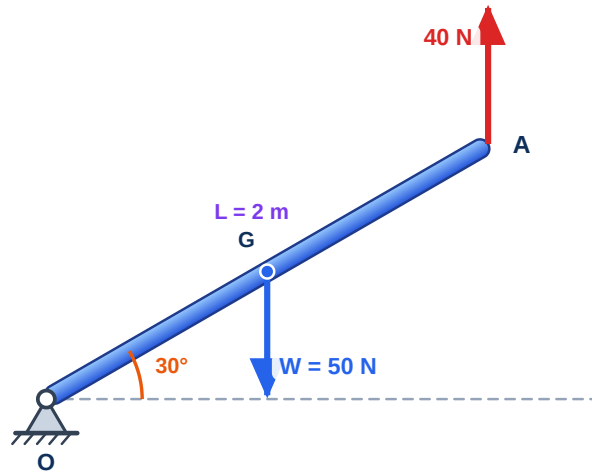
b) 90 N

c) 150 N

d) 120 N



A uniform rod OA of length 2 m and weight 50 N is pivoted at O and held at 30° above the horizontal. A vertical force of 40 N acts upwards at its end A:



Calculate the moment of the weight about O.



Calculate the moment of the 40 N force about O.



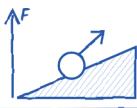
The net moment about O is

a) +26 N·m, the rod rises

b) -26 N·m, the rod falls

c) +113 N·m, the rod rises

d) zero



The graph shows how the moment of a constant force F about the hinge of a door changes with the angle θ between F and the door, when F always acts at the handle, 0.8 m from the hinge:



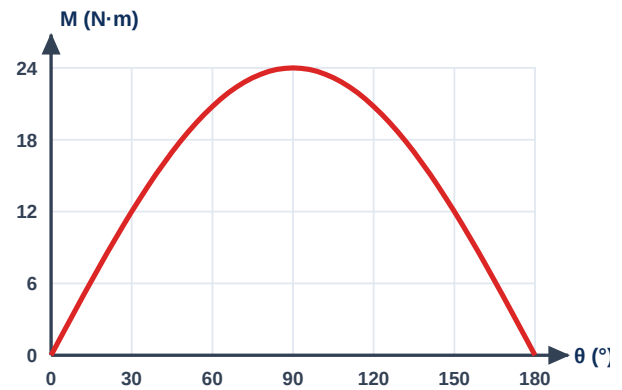
The magnitude of F is

- a) 19.2 N b) 24 N
c) 38.4 N d) 30 N



The moment when $\theta = 30^\circ$ is

- a) 6 N·m b) 12 N·m
c) 20.8 N·m d) 24 N·m



At which two angles does the moment equal 20.8 N·m?



A force of constant magnitude acts on a rod pivoted at O , always making an angle of 30° with the rod. The graph shows the moment of the force about O against the distance d between O and the point where the force acts:



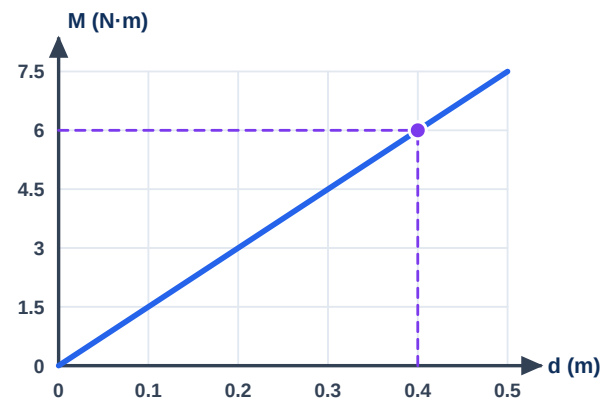
The slope of the line represents

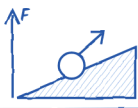
- a) F b) $F \sin \theta$
c) $F \cos \theta$ d) F / d



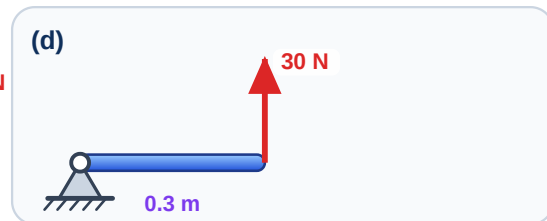
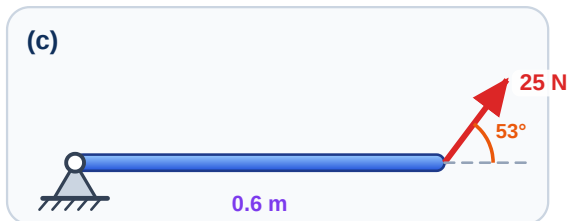
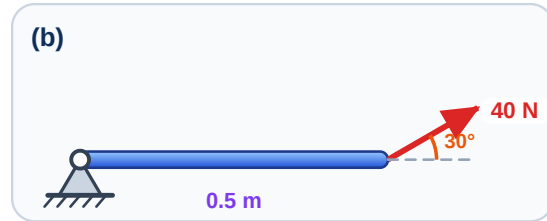
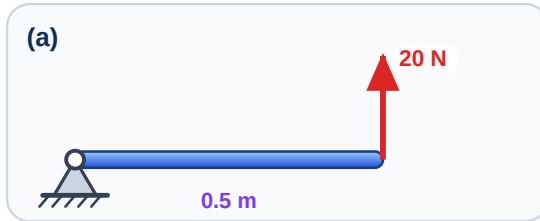
The magnitude of the force is

- a) 15 N b) 17.3 N
c) 7.5 N d) 30 N





In each of the four figures a force acts at the end of a light rod pivoted at one end. (Take $\sin 53^\circ = 0.8$)



The figure in which the force has the largest moment about the pivot is

a) (a)

b) (b)

c) (c)

d) (d)



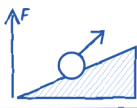
The two figures in which the forces have equal moments are

a) (a) and (d)

b) (b) and (c)

c) (c) and (d)

d) (a) and (b)



12

A cyclist pushes vertically downwards on a pedal with a constant force of 300 N. The crank that joins the pedal to the axle O is 17 cm long:

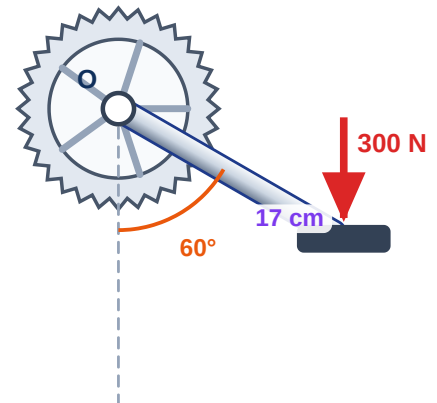


The maximum moment of this force about the axle is

- a) 25.5 N·m b) 44.2 N·m
c) 5100 N·m d) 51 N·m



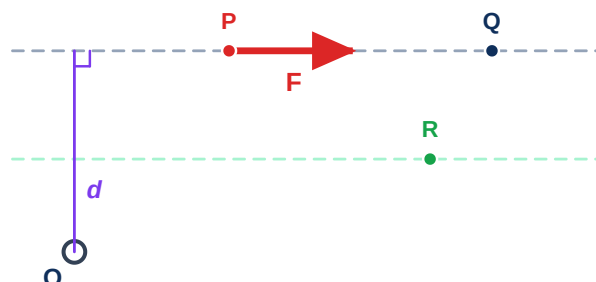
Calculate the moment when the crank makes 60° with the vertical, as in the figure.



In which position of the crank is the moment of this force zero? Why?

13

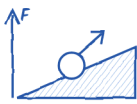
A force F acts at point P as shown. The dashed line is its line of action, and O is a fixed point:



If the force (same magnitude and direction) is moved to point Q on its line of action, its moment about O

- a) increases b) decreases c) does not change d) becomes zero





If the force is moved to point R on a parallel line nearer to O, its moment about O

a) increases

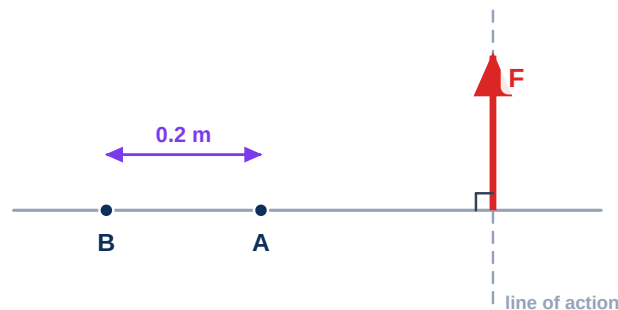
b) decreases

c) does not change

d) is reversed

14

A force F acts perpendicular to a straight line on which the points A and B lie, as shown. The moment of F about A is $12 \text{ N}\cdot\text{m}$, and its moment about B, which is 0.2 m farther from the line of action, is $20 \text{ N}\cdot\text{m}$:



The magnitude of F is

a) 40 N

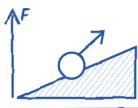
b) 60 N

c) 100 N

d) 160 N



Find the distance between A and the line of action of F .



15

ABCD is a square plate of side 0.4 m. Four forces act along its sides as shown: 10 N along AB, 20 N along BC, 30 N along CD and 40 N along DA:



The net moment of the four forces about A is

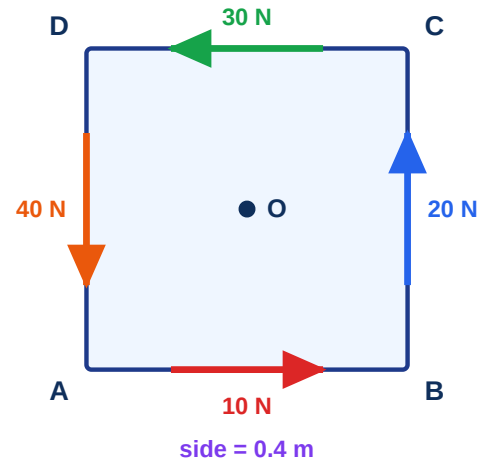
- a) +28 N·m b) +20 N·m
c) +40 N·m d) zero

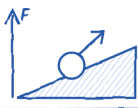


Calculate the net moment about B.



Calculate the net moment about the centre O of the plate.





16

The jib OB of a crane is 12 m long, pivoted at O and inclined at 53° to the horizontal. A load of weight 5000 N hangs from its end B. (Take $\sin 53^\circ = 0.8$, $\cos 53^\circ = 0.6$)



The moment of the load about O is

- a) $-4.8 \times 10^4 \text{ N}\cdot\text{m}$
- b) $-3.6 \times 10^4 \text{ N}\cdot\text{m}$
- c) $-6.0 \times 10^4 \text{ N}\cdot\text{m}$
- d) zero

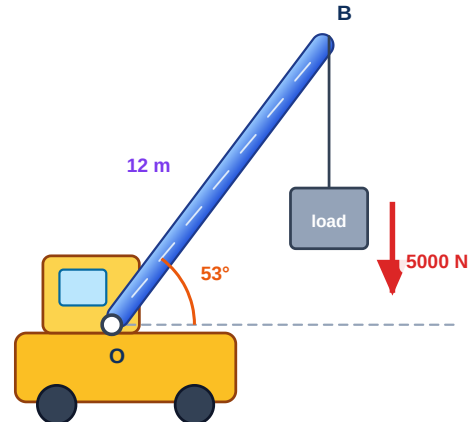


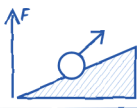
If the jib is lowered to 37° with the horizontal, the magnitude of the moment of the load

- a) increases to $4.8 \times 10^4 \text{ N}\cdot\text{m}$
- b) decreases to $2.4 \times 10^4 \text{ N}\cdot\text{m}$
- c) does not change
- d) becomes zero



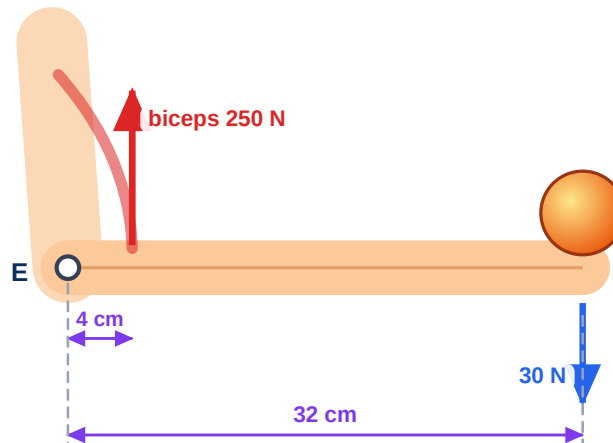
At what position of the jib is the moment of the load about O equal to zero?





17

The forearm is held horizontal. The biceps muscle pulls vertically upwards with a force of 250 N at a point 4 cm from the elbow E, and the hand holds a ball of weight 30 N at 32 cm from E. (Neglect the weight of the forearm.)



The moment of the ball's weight about the elbow is

a) $-9.6 \text{ N}\cdot\text{m}$

b) $+9.6 \text{ N}\cdot\text{m}$

c) $-0.96 \text{ N}\cdot\text{m}$

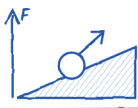
d) $-96 \text{ N}\cdot\text{m}$



Calculate the net moment about the elbow, and state what happens to the forearm.



Explain why the biceps must exert a force much larger than the weight of the ball.



18

A wheel of radius 0.3 m can rotate about its axle O. Four forces act on its rim as shown:



What is the moment of the 80 N force about O? Why?

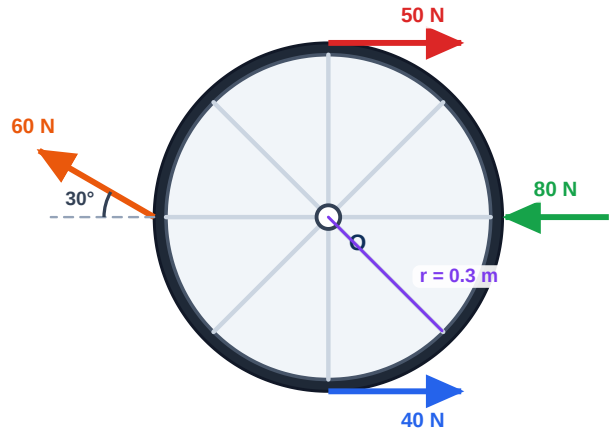


The moment of the 60 N force about O equals

- a) $-18 \text{ N}\cdot\text{m}$ b) $+9 \text{ N}\cdot\text{m}$
c) $-15.6 \text{ N}\cdot\text{m}$ d) $-9 \text{ N}\cdot\text{m}$

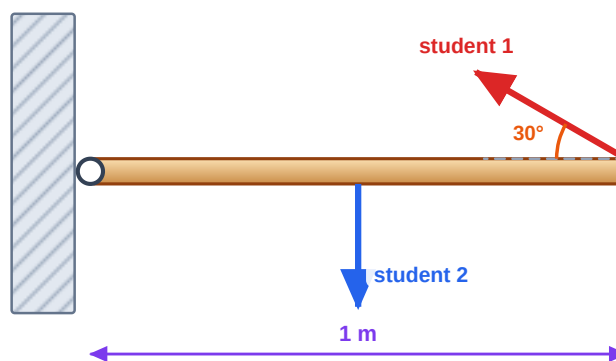


Calculate the net moment about O and the direction of rotation.



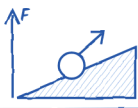
19

A door of width 1 m is hinged at one side. Student 1 pushes at the handle with a force making 30° with the door, while student 2 pushes perpendicular to the door at its middle, as shown:



If the two students produce moments of equal magnitude, the ratio F_1 / F_2 is

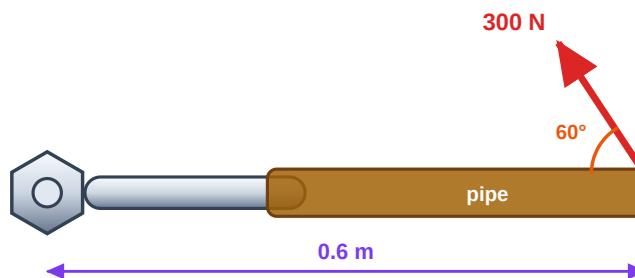
- a) $1/2$ b) 2 c) 1 d) 4



If $F_1 = 60 \text{ N}$ and $F_2 = 50 \text{ N}$, find the net moment about the hinge and state whose push wins.

20

A nut needs a moment of $120 \text{ N}\cdot\text{m}$ to loosen it. A mechanic can exert a force of at most 300 N :



The shortest spanner he can use (pushing perpendicular to it) has length

a) 0.25 m

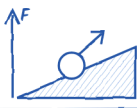
b) 0.6 m

c) 2.5 m

d) 0.4 m



He has only a 0.25 m spanner, so he slips a pipe over it to make its effective length 0.6 m . Because of the space around the nut, his 300 N force makes 60° with the handle, as shown. Will the nut be loosened? Show your calculation.



21

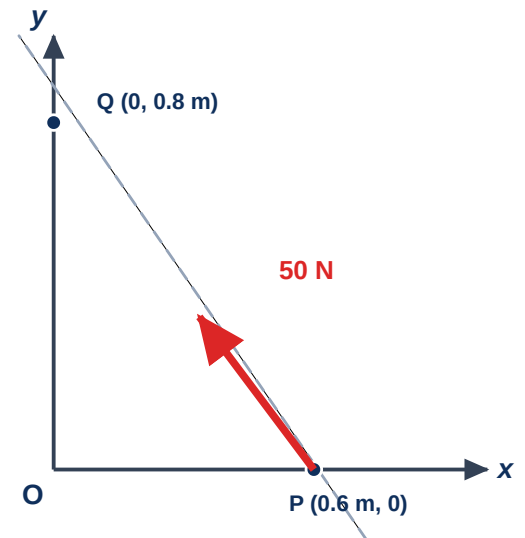
A force of 50 N acts along the line joining the points P (0.6 m, 0) and Q (0, 0.8 m), directed from P towards Q, as shown:

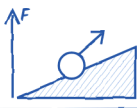


I The perpendicular distance from the origin O to the line of action is

- a) 0.6 m b) 0.8 m
c) 0.48 m d) 0.5 m

II Calculate the moment of the force about O, and check your answer by resolving the force at P.





22

A force whose horizontal component is 30 N and vertical component is 40 N acts at point A, which is 0.5 m to the right of and 0.2 m above the pivot O, as shown:

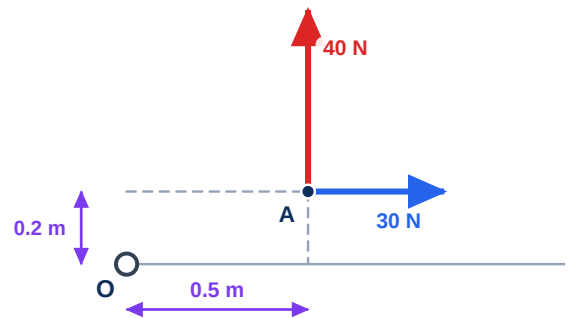


Calculate the moment of each component about O.



The moment of the force itself about O equals

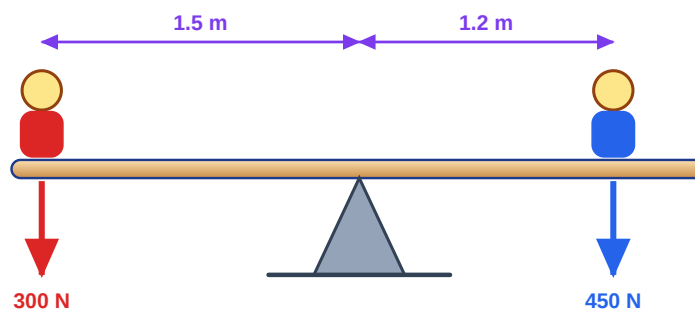
- a) +26 N·m b) +25 N·m
c) +14 N·m d) +20 N·m



Find the moment arm of the force about O.

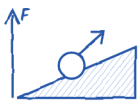
23

Two children sit on a light plank pivoted at its centre: a child of weight 300 N sits 1.5 m to the left of the pivot, and a child of weight 450 N sits 1.2 m to the right:



The net moment about the pivot is

- a) +90 N·m, anticlockwise b) zero
c) -990 N·m, clockwise d) -90 N·m, clockwise



Where should a third child of weight 200 N sit to make the net moment zero?



Without the third child, if the 450 N child moves 0.2 m towards the pivot, the plank

- a) turns clockwise
- b) turns anticlockwise
- c) does not start to rotate
- d) turns about the child

24

A rod OA of length 0.8 m is pivoted at O and makes 37° with the vertical. A constant horizontal force of 25 N acts at A as shown. (Take $\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$)



The moment of the force about O is

- a) -16 N·m
- b) -12 N·m
- c) +16 N·m
- d) -20 N·m

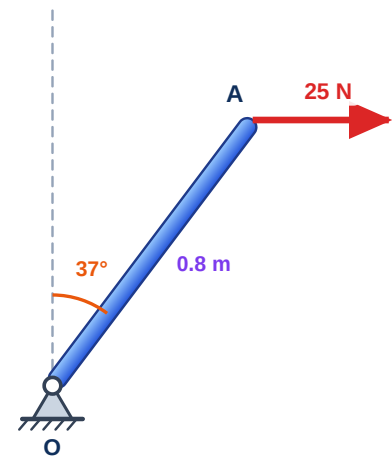


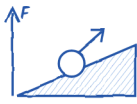
The moment of this force about O is largest when the rod is

- a) horizontal
- b) at 45°
- c) vertical
- d) at 37° with the horizontal



At what position of the rod is the moment of this force zero? Why?





25

Give the scientific reason for each of the following:



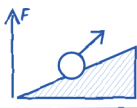
Door handles are placed at the edge of the door farthest from the hinges.



Pushing a door towards its hinges does not open it, however large the force is.



The moment of a force is measured in $\text{N}\cdot\text{m}$ but not in joules, although $1 \text{ N}\cdot\text{m} = 1 \text{ J}$.



26

A uniform horizontal pole AB of length 2 m and weight 40 N is fixed to a wall at A. A sign of weight 60 N hangs from its end B, and a cable attached at B makes 30° with the pole, as shown:



The total moment of the two weights about A is

- a) -100 N·m b) -200 N·m
c) -160 N·m d) -120 N·m

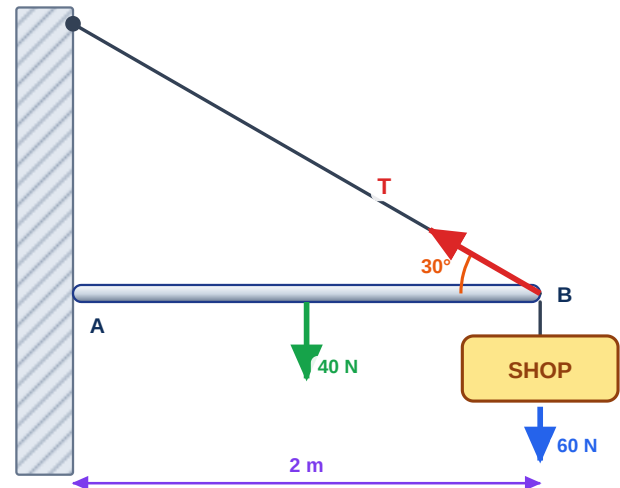


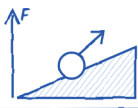
If the tension in the cable is 250 N, calculate the net moment about A.



The tension that makes the net moment about A equal to zero is

- a) 160 N b) 80 N
c) 320 N d) 185 N





27

A force $F = 20 \text{ N}$ acts at point P of a rod pivoted at O, where $OP = 0.5 \text{ m}$. The line of action of the force passes at a perpendicular distance of 0.3 m from O, as shown:

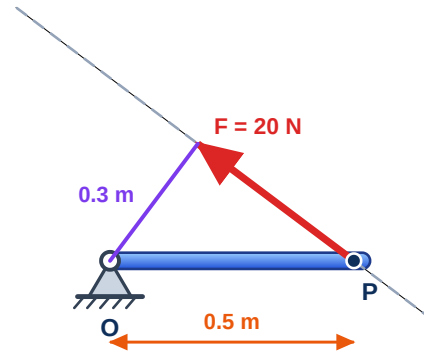


The moment arm of the force about O is

- a) 0.5 m b) 0.3 m
c) 0.4 m d) 0.8 m



Calculate the moment of the force about O and the angle between F and the rod.



28

Two forces F_1 and F_2 act on a rod pivoted at O. F_1 acts perpendicular to the rod at a distance d from O, while F_2 acts at a distance $3d$, making 30° with the rod:



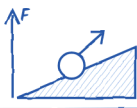
If the two moments are equal in magnitude, then F_1 / F_2 equals

- a) $1/3$ b) 3 c) $2/3$ d) $3/2$



If F_2 is turned (at the same point) to become perpendicular to the rod, its moment becomes

- a) half its value b) twice its value
c) three times its value d) unchanged



20

Choose the correct answer:

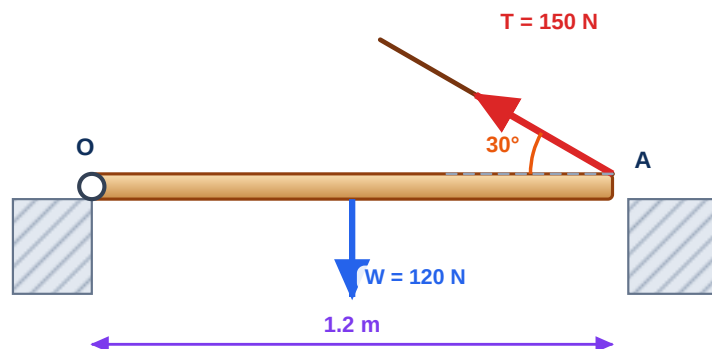


- I The force acting on a rod is doubled, the distance from the pivot to its point of application is doubled, and the angle between the force and the rod changes from 90° to 30° . The moment becomes
- a) four times b) twice c) the same d) half

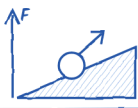
- II A force is halved while its angle with the rod changes from 30° to 90° (same point). Its moment
- a) is halved b) is doubled c) does not change d) is quartered

30

A uniform trapdoor OA of length 1.2 m and weight 120 N is hinged at O and held horizontal. A rope pulls its free end A with a tension of 150 N at 30° to the door, as shown:



- I Calculate the net moment about O, and state whether the trapdoor starts to rise or to fall.



If the rope were tied at the middle of the door instead (same tension, same angle), the trapdoor

- a) would start to rise b) would stay at rest
c) would start to fall d) would slide

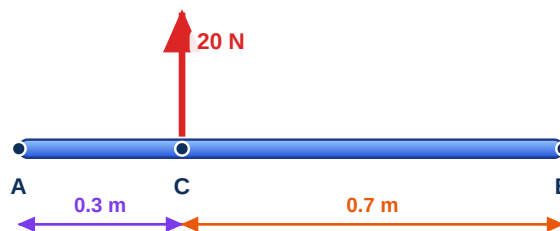


With the rope at A and at 30° , the smallest tension that just starts to raise the door is (just above)

- a) 60 N b) 144 N c) 240 N d) 120 N



A force of 20 N acts upwards, perpendicular to a rod AB of length 1 m, at point C where $AC = 0.3$ m, as shown:



The moment of the force about A is

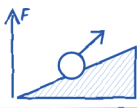
- a) +6 N·m b) -6 N·m c) +14 N·m d) -14 N·m



Calculate the moment of the same force about B and about C.



What do you conclude?



32

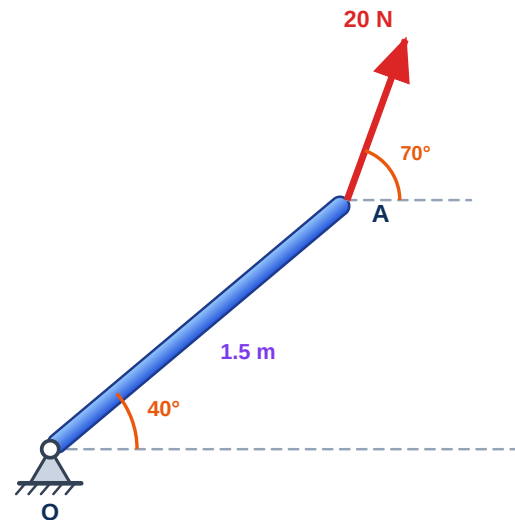
A rod OA of length 1.5 m is pivoted at O and makes 40° with the horizontal. A force of 20 N acts at A at 70° above the horizontal, as shown:



The moment of the force about O is

- a) $-15 \text{ N}\cdot\text{m}$ b) $+15 \text{ N}\cdot\text{m}$
c) $+26 \text{ N}\cdot\text{m}$ d) $+28.2 \text{ N}\cdot\text{m}$

The force keeps its magnitude but is turned to act at 10° above the horizontal. Find the new moment.



33

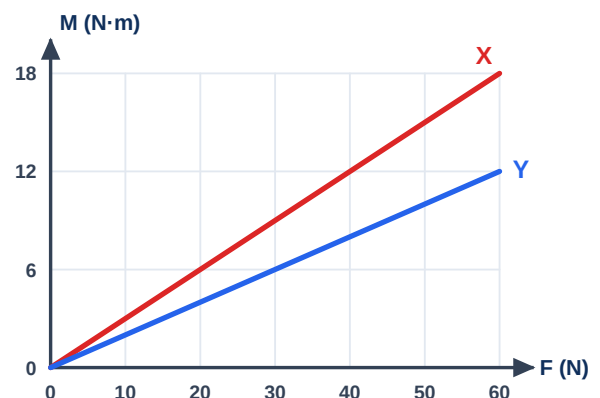
The graph shows the moment M against the force F for two spanners X and Y, when the force is always applied perpendicular to the end of each spanner:



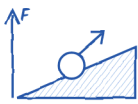
The longer spanner is

- a) X b) Y
c) both are equal d) it cannot be known

Find the length of each spanner.



What force is needed with spanner Y to produce a moment of $18 \text{ N}\cdot\text{m}$?



34

A force acting on a body pivoted at O produces a clockwise moment of 12 N·m:



If the force is reversed in direction (same line of action), its moment becomes

- a) +12 N·m b) -12 N·m c) zero d) -24 N·m

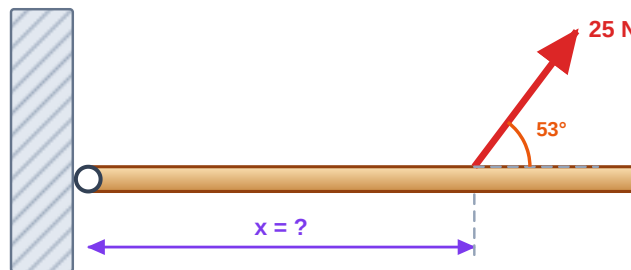


If instead the force is doubled and its line of action is moved parallel to itself halfway towards O, the moment becomes

- a) -24 N·m b) -6 N·m c) +12 N·m d) -12 N·m

35

A force of 25 N acts on a door at 53° to the plane of the door, at an unknown distance x from the hinge. Its moment about the hinge is 16 N·m. (Take $\sin 53^\circ = 0.8$)

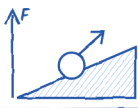


The distance x equals

- a) 0.64 m b) 1.07 m c) 0.8 m d) 0.6 m

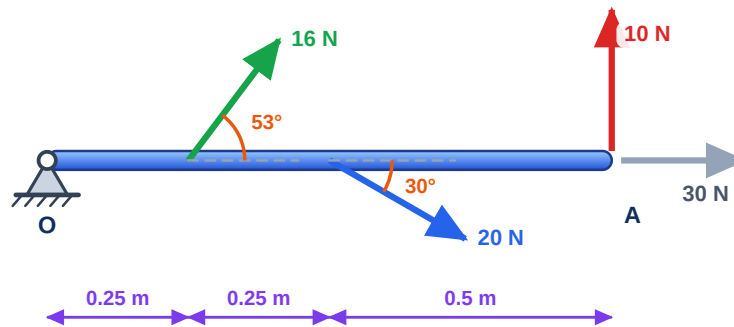


Find the force acting perpendicular to the door at the same point that gives the same moment.



36

Four forces act on a light rod OA of length 1 m pivoted at O, as shown: 16 N at 0.25 m making 53° with the rod, 20 N at the middle making 30° below the rod, 10 N at A perpendicular to the rod, and 30 N at A along the rod. (Take $\sin 53^\circ = 0.8$)



What is the moment of the 30 N force about O? Why?



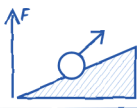
The net moment about O is

a) +18.2 N·m

b) +8.2 N·m

c) -8.2 N·m

d) +11.8 N·m



37

A ladder 5 m long rests with its foot O on the ground and its top against a smooth wall, making 53° with the ground. A man of weight 700 N stands 3 m up the ladder, and the wall pushes the top of the ladder horizontally with a force $R = 400$ N. (Take $\sin 53^\circ = 0.8$, $\cos 53^\circ = 0.6$)

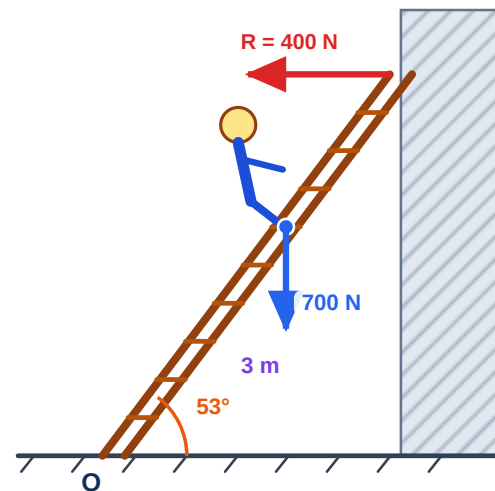


The moment of the man's weight about O is

- a) $-2100 \text{ N}\cdot\text{m}$ b) $-1680 \text{ N}\cdot\text{m}$
c) $-1260 \text{ N}\cdot\text{m}$ d) $-1470 \text{ N}\cdot\text{m}$



Calculate the moment of R about O.



As the man climbs higher, the moment of his weight about O

- a) increases b) decreases
c) does not change d) becomes zero

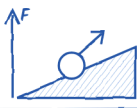
38

Choose the correct answer:



If the moment of a force about a point O is zero, then the force

- a) must be zero
b) is perpendicular to the arm
c) is zero or its line of action passes through O
d) must be very small



Which of the following does not change the moment of a force about a point?

- a) changing the magnitude of the force
- b) changing the angle between the force and the arm
- c) moving the point about which the moment is taken
- d) sliding the force along its line of action



The moment of a force about a point equals the force multiplied by

- a) the perpendicular distance between the point and the line of action
- b) the distance between the point and the point of application
- c) the length of the body
- d) the displacement of the body



The minute hand of a wall clock has a weight of 0.2 N acting at its midpoint, 6 cm from the centre of the clock. In the figure the hand points at the 10 o'clock mark (50 minutes):



The moment of the weight of the hand about the centre is largest when the hand points at

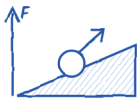
- a) 12
- b) 6
- c) 3 or 9
- d) 10



Calculate the moment of the weight about the centre in the position shown.



At which marks is this moment zero?



40

A uniform rod OA of length 1.2 m and weight 50 N is pivoted at O and makes 37° with the horizontal. A horizontal force F acts at its end A. (Take $\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$)

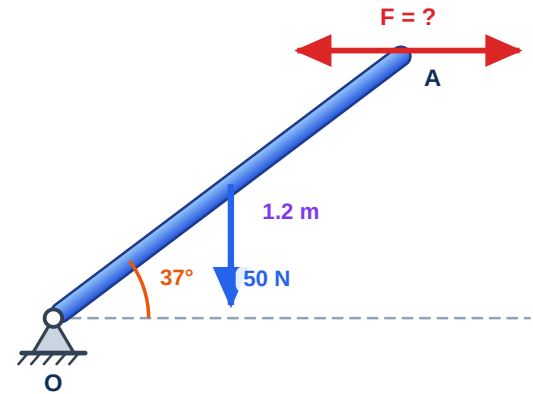


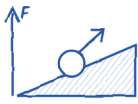
Calculate the moment of the weight about O.



To make the net moment about O equal to zero, the force F must be

- | | |
|------------------------|-----------------------|
| a) 33.3 N to the right | b) 41.7 N to the left |
| c) 25 N to the left | d) 33.3 N to the left |





Chapter 2 · Lesson 1

Model Answers

Every question solved step by step · anticlockwise (+) · clockwise (-)

1

- I The force is perpendicular to the spanner, so the moment arm is the whole length :

$$d = 25 \text{ cm} = 0.25 \text{ m}$$

$$M = F \times d = 80 \times 0.25 = 20 \text{ N}\cdot\text{m}$$

b) 20 N·m

- II $d = 0.25 / 2 = 0.125 \text{ m}$

$$M = F \times d = 80 \times 0.125 = 10 \text{ N}\cdot\text{m}$$

$M \propto d$: halving the arm halves the moment.

d) 10 N·m

2

- I F_2 acts along the door, towards the hinge.
Its line of action passes through the hinge, so its moment arm $d = 0$.
 $M = F \times d = F_2 \times 0 = 0$

F_2 : its line of action passes through the hinge

- II $M_1 = 30 \times 0.9 = 27 \text{ N}\cdot\text{m}$
 $M_3 = 30 \times 0.45 = 13.5 \text{ N}\cdot\text{m}$
 $M_1 : M_3 = 27 : 13.5 = 2 : 1$

b) 2 : 1

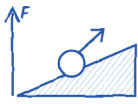
3

- I $d = L \sin \theta = 0.5 \times \sin 30^\circ$
 $d = 0.5 \times 0.5 = 0.25 \text{ m}$

d) 0.25 m

- II $M = F L \sin \theta = 40 \times 0.5 \times \sin 30^\circ = 10 \text{ N}\cdot\text{m}$
The force turns the rod anticlockwise, so the moment is positive : $M = +10 \text{ N}\cdot\text{m}$

a) +10 N·m



4

I $M = F \times d \rightarrow \text{newton} \times \text{metre} = \text{N}\cdot\text{m}$

d) $\text{N}\cdot\text{m}$

II If the line of action passes through the point, the moment arm $d = 0$, so $M = F \times 0 = 0$ whatever the size of the force.

b) the line of action of the force passes through the point

III $M = F \times d$: for the same moment, a larger arm d needs a smaller force F .

c) increase the moment arm, so a smaller force gives the needed moment

5

I $M = F \times d = 30 \times 0.6 = 18 \text{ N}\cdot\text{m}$

B is to the right of O and the force is downwards $\rightarrow$ clockwise

$M = -18 \text{ N}\cdot\text{m}$

c) $-18 \text{ N}\cdot\text{m}$

II 20 N at A : $+20 \times 0.6 = +12 \text{ N}\cdot\text{m}$ (anticlockwise)

30 N at B : $-30 \times 0.6 = -18 \text{ N}\cdot\text{m}$ (clockwise)

25 N at C : $+25 \times 0.3 = +7.5 \text{ N}\cdot\text{m}$ (anticlockwise)

$\Sigma M = 12 - 18 + 7.5 = +1.5 \text{ N}\cdot\text{m}$

$\Sigma M = +1.5 \text{ N}\cdot\text{m} \rightarrow$ the rod starts to rotate anticlockwise

6

I $M = F r \sin \theta \rightarrow \sin \theta = M / (F r) = 15 / (60 \times 0.5) = 0.5$

$\sin \theta = 0.5$ has two answers between 0° and 180° :

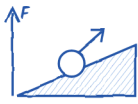
$\theta = 30^\circ$ or $\theta = 150^\circ$

d) 30° or 150°

II $M = F r \sin \theta$ is largest for a given F when $\sin \theta = 1$ ($\theta = 90^\circ$)

$F_{\min} = M / r = 15 / 0.5 = 30 \text{ N}$

$F_{\min} = 30 \text{ N}$, perpendicular to OP



7

I

$$OP = \sqrt{(0.3^2 + 0.4^2)} = \sqrt{0.25} = 0.5 \text{ m}$$

The largest moment arm possible is OP itself, reached when $F \perp OP$

$$F_{\min} = M / OP = 36 / 0.5 = 72 \text{ N}$$

$F_{\min} = 72 \text{ N}$, perpendicular to OP

II

For a horizontal force the moment arm is the vertical distance of P from O : $d = 0.4 \text{ m}$

$$F = M / d = 36 / 0.4 = 90 \text{ N}$$

a) 90 N

III

For a vertical force the moment arm is the horizontal distance of P from O : $d = 0.3 \text{ m}$

$$F = M / d = 36 / 0.3 = 120 \text{ N}$$

d) 120 N

8

I

The weight acts at the midpoint G, 1 m from O along the rod
moment arm = horizontal distance of G = $1 \times \cos 30^\circ = 0.866 \text{ m}$
 $M_W = -50 \times 0.866 = -43.3 \text{ N}\cdot\text{m}$ (clockwise)

$M_W = -43.3 \text{ N}\cdot\text{m}$

II

moment arm = horizontal distance of A = $2 \times \cos 30^\circ = 1.732 \text{ m}$
 $M = +40 \times 1.732 = +69.3 \text{ N}\cdot\text{m}$ (anticlockwise)

$M = +69.3 \text{ N}\cdot\text{m}$

III

$\Sigma M = +69.3 - 43.3 = +26 \text{ N}\cdot\text{m}$
positive $\rightarrow$ anticlockwise $\rightarrow$ the rod starts to rise

a) $+26 \text{ N}\cdot\text{m}$, the rod rises

9

I

The maximum moment (at $\theta = 90^\circ$) is $M_{\max} = F \times d$
 $F = M_{\max} / d = 24 / 0.8 = 30 \text{ N}$

d) 30 N

II

$$M = F d \sin \theta = 24 \times \sin 30^\circ = 12 \text{ N}\cdot\text{m}$$

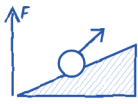
b) 12 N·m

III

$$\sin \theta = M / M_{\max} = 20.8 / 24 = 0.866$$

$$\theta = 60^\circ \text{ or } \theta = 180^\circ - 60^\circ = 120^\circ$$

$\theta = 60^\circ$ and $\theta = 120^\circ$



10

- I $M = F d \sin \theta = (F \sin \theta) \times d$
M is proportional to d, and the constant of proportionality (the slope) is $F \sin \theta$
b) $F \sin \theta$
- II slope = $6 / 0.4 = 15 \text{ N}$
 $F \sin 30^\circ = 15 \rightarrow F = 15 / 0.5 = 30 \text{ N}$
d) 30 N

11

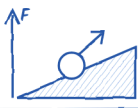
- I (a) $M = 20 \times 0.5 = 10 \text{ N}\cdot\text{m}$
(b) $M = 40 \times 0.5 \times \sin 30^\circ = 10 \text{ N}\cdot\text{m}$
(c) $M = 25 \times 0.6 \times \sin 53^\circ = 25 \times 0.6 \times 0.8 = 12 \text{ N}\cdot\text{m}$
(d) $M = 30 \times 0.3 = 9 \text{ N}\cdot\text{m}$
The largest is (c), although (b) has the largest force.
c) (c)
- II From part I : (a) and (b) both give $10 \text{ N}\cdot\text{m}$.
d) (a) and (b)

12

- I The moment arm is largest ($= 0.17 \text{ m}$) when the crank is horizontal
 $M_{\text{max}} = 300 \times 0.17 = 51 \text{ N}\cdot\text{m}$
d) 51 N·m
- II The force is vertical, so the angle between F and the crank = 60°
 $M = F L \sin \theta = 300 \times 0.17 \times \sin 60^\circ = 44.2 \text{ N}\cdot\text{m}$
M $\approx$ 44.2 N·m
- III When the crank is vertical (pedal at the top or at the bottom),
the vertical force acts along the crank and its line of action passes through O $\rightarrow$
 $d = 0$
crank vertical $\rightarrow M = 0$

13

- I $M = F \times d$, where d is the perpendicular distance from O to the line of action
Moving the force along its own line of action does not change d, so M stays the same.
c) does not change
- II The new line of action is nearer to O, so d is smaller $\rightarrow M = F d$ decreases.
b) decreases



14

- I $M_A = F d_A$ and $M_B = F (d_A + 0.2)$
 $M_B - M_A = F \times 0.2$
 $20 - 12 = 0.2 F \rightarrow F = 40 \text{ N}$

a) 40 N

- II $d_A = M_A / F = 12 / 40 = 0.3 \text{ m}$
 $d_A = 0.3 \text{ m}$

15

- I 10 N and 40 N act along lines through A $\rightarrow$ zero moment
 20 N (up the side BC): $+20 \times 0.4 = +8 \text{ N}\cdot\text{m}$
 30 N (along CD, 0.4 m above A): $+30 \times 0.4 = +12 \text{ N}\cdot\text{m}$
 $\Sigma M_A = 8 + 12 = +20 \text{ N}\cdot\text{m}$ (anticlockwise)

b) +20 N·m

- II 10 N and 20 N act along lines through B $\rightarrow$ zero moment
 30 N along CD, 0.4 m above B: $+30 \times 0.4 = +12 \text{ N}\cdot\text{m}$
 40 N along DA, 0.4 m to the left of B: $+40 \times 0.4 = +16 \text{ N}\cdot\text{m}$
 $\Sigma M_B = 12 + 16 = +28 \text{ N}\cdot\text{m}$

$\Sigma M_B = +28 \text{ N}\cdot\text{m}$

- III Each force is 0.2 m from O, and all of them turn the plate anticlockwise
 $\Sigma M_O = 0.2 \times (10 + 20 + 30 + 40) = +20 \text{ N}\cdot\text{m}$
 The same forces give different net moments about different points.

$\Sigma M_O = +20 \text{ N}\cdot\text{m}$

16

- I The weight is vertical, so its moment arm is the horizontal distance of B from O
 $d = 12 \cos 53^\circ = 12 \times 0.6 = 7.2 \text{ m}$
 $M = -5000 \times 7.2 = -3.6 \times 10^4 \text{ N}\cdot\text{m}$ (clockwise)

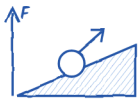
b) $-3.6 \times 10^4 \text{ N}\cdot\text{m}$

- II $d = 12 \cos 37^\circ = 12 \times 0.8 = 9.6 \text{ m}$
 $|M| = 5000 \times 9.6 = 4.8 \times 10^4 \text{ N}\cdot\text{m} \rightarrow$ it increases

a) increases to $4.8 \times 10^4 \text{ N}\cdot\text{m}$

- III When the jib is vertical, B is directly above O,
 so the weight's line of action passes through O $\rightarrow d = 0$

jib vertical (90° with the horizontal)



17

I $M = -30 \times 0.32 = -9.6 \text{ N}\cdot\text{m}$ (clockwise)

a) $-9.6 \text{ N}\cdot\text{m}$

II biceps : $M = +250 \times 0.04 = +10 \text{ N}\cdot\text{m}$ (anticlockwise)

$\Sigma M = +10 - 9.6 = +0.4 \text{ N}\cdot\text{m}$

$\Sigma M = +0.4 \text{ N}\cdot\text{m} \rightarrow$ the forearm starts to rise

III Its moment arm (4 cm) is 8 times shorter than that of the ball (32 cm), so to give a comparable moment its force must be about 8 times larger.

the moment arm of the muscle is very short

18

I The 80 N force acts along the radius, towards O, so its line of action passes through O $\rightarrow d = 0 \rightarrow M = 0$

zero

II Only the component perpendicular to the radius turns the wheel :

$F_{\perp} = 60 \sin 30^{\circ} = 30 \text{ N}$ (vertically upwards at the left point)

$M = -30 \times 0.3 = -9 \text{ N}\cdot\text{m}$ (clockwise)

d) $-9 \text{ N}\cdot\text{m}$

III 50 N at the top, to the right : $-50 \times 0.3 = -15 \text{ N}\cdot\text{m}$

40 N at the bottom, to the right : $+40 \times 0.3 = +12 \text{ N}\cdot\text{m}$

80 N : 0 60 N : $-9 \text{ N}\cdot\text{m}$

$\Sigma M = -15 + 12 + 0 - 9 = -12 \text{ N}\cdot\text{m}$

$\Sigma M = -12 \text{ N}\cdot\text{m} \rightarrow$ clockwise

19

I $M_1 = F_1 \times 1 \times \sin 30^{\circ} = 0.5 F_1$

$M_2 = F_2 \times 0.5$

$0.5 F_1 = 0.5 F_2 \rightarrow F_1 / F_2 = 1$

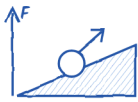
c) 1

II $M_1 = +60 \times 1 \times 0.5 = +30 \text{ N}\cdot\text{m}$

$M_2 = -50 \times 0.5 = -25 \text{ N}\cdot\text{m}$

$\Sigma M = +5 \text{ N}\cdot\text{m}$

$\Sigma M = +5 \text{ N}\cdot\text{m} \rightarrow$ the door turns in the direction of student 1



20

I $d_{\min} = M / F_{\max} = 120 / 300 = 0.4 \text{ m}$

d) 0.4 m

II $M = F L \sin \theta = 300 \times 0.6 \times \sin 60^\circ$

$M = 300 \times 0.6 \times 0.866 = 155.9 \text{ N}\cdot\text{m}$

$155.9 \text{ N}\cdot\text{m} > 120 \text{ N}\cdot\text{m}$

yes : $M \approx 156 \text{ N}\cdot\text{m}$, more than the 120 N·m needed

21

I OPQ is a right-angled triangle with legs 0.6 m and 0.8 m, so PQ = 1.0 m

twice its area = $0.6 \times 0.8 = \text{PQ} \times d$

$d = 0.48 / 1.0 = 0.48 \text{ m}$

c) 0.48 m

II $M = F d = 50 \times 0.48 = 24 \text{ N}\cdot\text{m}$, anticlockwise $\rightarrow +24 \text{ N}\cdot\text{m}$

check : at P the force has components 30 N (horizontal) and 40 N (vertical)

the horizontal component lies along the x-axis, through O $\rightarrow$ no moment

the vertical component : $M = +40 \times 0.6 = +24 \text{ N}\cdot\text{m} \checkmark$

$M = +24 \text{ N}\cdot\text{m}$

22

I vertical 40 N : arm = 0.5 m $\rightarrow M = +40 \times 0.5 = +20 \text{ N}\cdot\text{m}$

horizontal 30 N (to the right, above O) : arm = 0.2 m $\rightarrow M = -30 \times 0.2 = -6 \text{ N}\cdot\text{m}$

+20 N·m and -6 N·m

II The moment of a force = the sum of the moments of its components

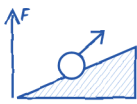
$M = +20 - 6 = +14 \text{ N}\cdot\text{m}$

c) +14 N·m

III $F = \sqrt{(30^2 + 40^2)} = 50 \text{ N}$

$d = M / F = 14 / 50 = 0.28 \text{ m}$

d = 0.28 m



23

- I left child : $+300 \times 1.5 = +450 \text{ N}\cdot\text{m}$
right child : $-450 \times 1.2 = -540 \text{ N}\cdot\text{m}$
 $\Sigma M = 450 - 540 = -90 \text{ N}\cdot\text{m} \rightarrow \text{clockwise}$

d) $-90 \text{ N}\cdot\text{m}$, clockwise

- II an extra $+90 \text{ N}\cdot\text{m}$ is needed $\rightarrow$ the child sits on the left
 $200 \times d = 90 \rightarrow d = 0.45 \text{ m}$

0.45 m to the left of the pivot

- III right child : $-450 \times 1.0 = -450 \text{ N}\cdot\text{m}$
 $\Sigma M = +450 - 450 = 0 \rightarrow \text{no rotation}$

c) does not start to rotate

24

- I For a horizontal force, the moment arm is the vertical height of A above O
 $d = 0.8 \cos 37^\circ = 0.8 \times 0.8 = 0.64 \text{ m}$
 $M = -25 \times 0.64 = -16 \text{ N}\cdot\text{m}$ (clockwise)

a) $-16 \text{ N}\cdot\text{m}$

- II The arm (the height of A) is largest when the rod is vertical : $d = 0.8 \text{ m}$
 $M_{\text{max}} = 25 \times 0.8 = 20 \text{ N}\cdot\text{m}$

c) vertical

- III When the rod is horizontal, the horizontal force acts along the rod,
its line of action passes through O $\rightarrow d = 0$

rod horizontal $\rightarrow M = 0$

25

- I $M = F \times d$: the largest moment arm gives the needed moment with the smallest force.

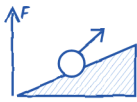
largest moment arm $\rightarrow$ smallest force needed

- II The line of action of the force passes through the axis of rotation (the hinges),
so $d = 0$ and $M = F \times 0 = 0$.

the moment arm is zero

- III The joule is the unit of work and energy, where the distance is along the displacement.
In the moment, the distance is perpendicular to the force and nothing is displaced by it;
the moment measures a turning effect (a vector), not energy.

the moment is a turning effect, not work or energy



26

- I pole weight at its middle : $-40 \times 1 = -40 \text{ N}\cdot\text{m}$
 sign at B : $-60 \times 2 = -120 \text{ N}\cdot\text{m}$
 total = $-160 \text{ N}\cdot\text{m}$ (clockwise)

c) $-160 \text{ N}\cdot\text{m}$

- II moment of T : $+T \times 2 \times \sin 30^\circ = +250 \times 2 \times 0.5 = +250 \text{ N}\cdot\text{m}$
 $\Sigma M = +250 - 160 = +90 \text{ N}\cdot\text{m}$

$\Sigma M = +90 \text{ N}\cdot\text{m}$

- III $T \times 2 \times \sin 30^\circ = 160$
 $T \times 1 = 160 \rightarrow T = 160 \text{ N}$

a) 160 N

27

- I The moment arm is the perpendicular distance from O to the line of action, not the distance OP : $d = 0.3 \text{ m}$

b) 0.3 m

- II $M = F \times d = 20 \times 0.3 = 6 \text{ N}\cdot\text{m}$, anticlockwise $\rightarrow +6 \text{ N}\cdot\text{m}$
 $d = OP \sin \theta \rightarrow \sin \theta = 0.3 / 0.5 = 0.6 \rightarrow \theta = 37^\circ$

$M = +6 \text{ N}\cdot\text{m}$, $\theta = 37^\circ$

28

- I $F_1 \times d = F_2 \times 3d \times \sin 30^\circ$
 $F_1 = 1.5 F_2 \rightarrow F_1 / F_2 = 3/2$

d) $3/2$

- II $M \propto \sin \theta : \sin 90^\circ / \sin 30^\circ = 1 / 0.5 = 2$

b) twice its value

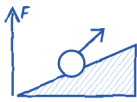
29

- I $M = F L \sin \theta$
 $M' / M = (2 \times 2 \times \sin 30^\circ) / (1 \times 1 \times \sin 90^\circ) = 4 \times 0.5 = 2$

b) twice

- II $M' / M = (0.5 \times \sin 90^\circ) / (1 \times \sin 30^\circ) = 0.5 / 0.5 = 1$

c) does not change



30

- I weight at the middle : $-120 \times 0.6 = -72 \text{ N}\cdot\text{m}$
 rope at A : $+150 \times 1.2 \times \sin 30^\circ = +90 \text{ N}\cdot\text{m}$
 $\Sigma M = -72 + 90 = +18 \text{ N}\cdot\text{m} \rightarrow$ anticlockwise

$\Sigma M = +18 \text{ N}\cdot\text{m} \rightarrow$ the trapdoor starts to rise

- II rope : $+150 \times 0.6 \times 0.5 = +45 \text{ N}\cdot\text{m}$
 $\Sigma M = -72 + 45 = -27 \text{ N}\cdot\text{m} \rightarrow$ clockwise $\rightarrow$ it falls

c) would start to fall

- III $T \times 1.2 \times \sin 30^\circ = 72$
 $0.6 T = 72 \rightarrow T = 120 \text{ N}$

d) 120 N

31

- I C is 0.3 m to the right of A and the force is upwards $\rightarrow$ anticlockwise
 $M_A = +20 \times 0.3 = +6 \text{ N}\cdot\text{m}$

a) $+6 \text{ N}\cdot\text{m}$

- II about B : C is 0.7 m to the left of B, force upwards $\rightarrow$ clockwise
 $M_B = -20 \times 0.7 = -14 \text{ N}\cdot\text{m}$
 about C : the force passes through C $\rightarrow M_C = 0$

$M_B = -14 \text{ N}\cdot\text{m}, M_C = 0$

- III The moment of a force depends on the point it is taken about :
 its value and even its sign can change.

the moment depends on the chosen point

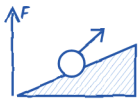
32

- I The angle between the force and the rod $= 70^\circ - 40^\circ = 30^\circ$
 $M = F L \sin \theta = 20 \times 1.5 \times \sin 30^\circ = 15 \text{ N}\cdot\text{m}$
 The force is turned anticlockwise relative to the rod $\rightarrow M = +15 \text{ N}\cdot\text{m}$

b) $+15 \text{ N}\cdot\text{m}$

- II The angle between the force and the rod $= 40^\circ - 10^\circ = 30^\circ$, but now on the other side of the rod
 $M = 20 \times 1.5 \times \sin 30^\circ = 15 \text{ N}\cdot\text{m}$, clockwise

$M = -15 \text{ N}\cdot\text{m}$



33

- I $M = F d \rightarrow$ the slope of each line equals the length d of the spanner
X has the steeper line $\rightarrow$ X is longer

a) X

- II X: $d = 18 / 60 = 0.3$ m
Y: $d = 12 / 60 = 0.2$ m

X = 0.3 m, Y = 0.2 m

- III $F = M / d = 18 / 0.2 = 90$ N

F = 90 N

34

- I Reversing the force reverses the sense of rotation:
-12 N·m (clockwise) $\rightarrow$ +12 N·m (anticlockwise)

a) +12 N·m

- II $M' = (2F) \times (d / 2) = F d$
The moment is unchanged: -12 N·m

d) -12 N·m

35

- I $M = F \times \sin \theta$
 $16 = 25 \times x \times 0.8 \rightarrow x = 16 / 20 = 0.8$ m

c) 0.8 m

- II $F_{\perp} \times 0.8 = 16 \rightarrow F_{\perp} = 20$ N
(it equals the perpendicular component of the 25 N force: $25 \times 0.8 = 20$ N)

$F_{\perp} = 20$ N

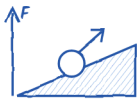
36

- I It acts along the rod, so its line of action passes through O $\rightarrow d = 0 \rightarrow M = 0$

zero

- II 16 N: $+16 \times 0.25 \times \sin 53^{\circ} = +16 \times 0.25 \times 0.8 = +3.2$ N·m
20 N: $-20 \times 0.5 \times \sin 30^{\circ} = -5$ N·m (it turns the rod clockwise)
10 N: $+10 \times 1 = +10$ N·m 30 N: 0
 $\Sigma M = 3.2 - 5 + 10 + 0 = +8.2$ N·m (anticlockwise)

b) +8.2 N·m



37

- I moment arm = horizontal distance of the man from O = $3 \cos 53^\circ = 1.8 \text{ m}$
 $M = -700 \times 1.8 = -1260 \text{ N}\cdot\text{m}$ (clockwise)

c) $-1260 \text{ N}\cdot\text{m}$

- II R is horizontal, so its arm is the height of the top of the ladder : $5 \sin 53^\circ = 4 \text{ m}$
 $M = +400 \times 4 = +1600 \text{ N}\cdot\text{m}$ (anticlockwise)

$M_R = +1600 \text{ N}\cdot\text{m}$

- III His horizontal distance from O increases as he climbs $\rightarrow$ the moment arm increases.

a) increases

38

- I $M = F d = 0 \rightarrow$ either $F = 0$ or $d = 0$ (line of action through O).

c) is zero or its line of action passes through O

- II Sliding along the line of action keeps the perpendicular distance d the same.

d) sliding the force along its line of action

- III This is the definition : $M = F \times d$, with d the perpendicular distance.

a) the perpendicular distance between the point and the line of action

39

- I The weight is vertical, so its arm is the horizontal distance of the midpoint from the centre,
 largest (= 6 cm) when the hand is horizontal : at 3 or 9

$$M_{\max} = 0.2 \times 0.06 = 1.2 \times 10^{-2} \text{ N}\cdot\text{m}$$

c) 3 or 9

- II The hand makes 60° with the vertical
 arm = $0.06 \times \sin 60^\circ = 0.052 \text{ m}$

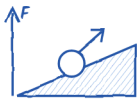
$$M = 0.2 \times 0.052 = 1.04 \times 10^{-2} \text{ N}\cdot\text{m}$$

the weight is to the left of the centre $\rightarrow$ anticlockwise

$M = +1.04 \times 10^{-2} \text{ N}\cdot\text{m}$

- III When the hand is vertical (at 12 or 6), the weight acts along the hand, through the centre.

at 12 and at 6



40

I arm = horizontal distance of the midpoint = $0.6 \cos 37^\circ = 0.6 \times 0.8 = 0.48 \text{ m}$
 $M_W = -50 \times 0.48 = -24 \text{ N}\cdot\text{m}$ (clockwise)

$M_W = -24 \text{ N}\cdot\text{m}$

II A is $1.2 \sin 37^\circ = 0.72 \text{ m}$ above O
 a horizontal force to the right at A turns the rod clockwise (like the weight),
 so F must point to the left to give an anticlockwise moment
 $F \times 0.72 = 24 \rightarrow F = 33.3 \text{ N}$ to the left

d) 33.3 N to the left